

This project is a machine learning-based hiring prediction system designed to evaluate candidate profiles and predict whether a recruiter will "Hire" or "Reject" them. Although the introductory markdown cell labels it an "Interactive Job Posting Fraud Predictor," the underlying Python code and dataset (AI-Based Hiring Prediction System) are explicitly built for candidate screening.

Key components of the project include:

- **Data Processing:** The system processes both textual candidate attributes (Skills, Education, Certifications, Job Role) and numerical attributes (Years of Experience, Salary Expectation, Project Count). The text fields are combined and cleaned for natural language processing.
- **Machine Learning Pipeline:** It utilizes a Scikit-Learn Pipeline equipped with a ColumnTransformer. This transformer applies TF-IDF vectorization to extract features from the combined text and uses a StandardScaler to normalize the numerical data.
- **Classification Model:** The core predictive model is a Logistic Regression classifier, which demonstrates high performance with a 97% accuracy rate in distinguishing between hired and rejected candidates during evaluation.
- **Interactive Interface:** The notebook concludes with a widget-based UI built using ipywidgets. It allows users to manually input a new candidate's text and numerical details via text boxes and sliders, click a "Predict Decision" button, and receive a real-time "HIRED (1)" or "REJECTED (0)" prediction.